



Self-Supervised Vibration Analytics for Predictive Maintenance of Multistage Compressors

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Abstract

Purpose Multistage compressors are vital industrial assets, and their continuous operation makes timely fault detection crucial to avoid unexpected downtime. Traditional vibration-based predictive maintenance methods largely depend on supervised learning techniques, which require extensive labeled fault data—data that are rarely available in real-world industrial environments. To address this limitation, this study presents a self-supervised learning (SSL) framework for vibration-based condition monitoring of multistage compressors, allowing fault-relevant features to be learned directly from unlabeled data.

Methods The proposed method employs a contrastive Siamese convolutional architecture with domain-specific temporal augmentations, followed by a lightweight supervised classifier trained using only a small number of labeled samples. Furthermore, an Entropy Divergence Rate (EDR) metric is introduced to capture distributional shifts in the learned feature space, enabling early detection of anomalous behavior. The framework is validated using a six-month in-house multistage compressor dataset along with an adapted Case Western Reserve University bearing dataset.

Results The results show an F1-score of 0.93 and a 28% improvement in early fault detection compared with FFT-based and fully supervised deep learning approaches.

Conclusion Overall, the proposed framework substantially reduces reliance on labeled data while enhancing early fault recognition, making it well suited for scalable and practical industrial predictive maintenance applications.

Keywords Self-supervised learning · Predictive maintenance · Vibration analysis · Multistage compressors · Fault detection · Condition monitoring

Introduction

Multistage compressors are essential components in a wide range of industrial applications, including oil and gas processing, chemical manufacturing, refrigeration, and air separation systems. Their complex multi-stage rotating assemblies and continuous operation under high mechanical

and thermal stresses make them susceptible to progressive degradation mechanisms such as bearing wear, rotor imbalance, and shaft misalignment. When left undetected, these incipient faults can propagate rapidly, leading to catastrophic failures, substantial economic losses, and serious safety hazards.

Conventional maintenance strategies, namely reactive and preventive maintenance, are increasingly insufficient in modern industrial settings. Reactive maintenance often results in unplanned shutdowns and production losses, whereas preventive maintenance may lead to the premature replacement of components that are still in healthy operating condition. As a result, predictive maintenance (PdM) has gained considerable attention, as it aims to anticipate failures through continuous condition monitoring and timely intervention. Among various PdM techniques, vibration analysis remains one of the most widely adopted approaches, owing to its high sensitivity to mechanical defects and its ability to capture early-stage fault signatures.

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With the advancement of machine learning, automated vibration-based fault diagnosis has become an active area of research. Most existing approaches, however, rely on supervised learning and require large quantities of labeled fault data. In practical industrial environments, such data are often scarce, costly to obtain, and incomplete, which significantly limits the deployment of supervised models. Although deep learning architectures such as convolutional neural networks (CNNs), and recurrent neural networks (RNNs) have improved feature extraction capabilities, their performance remains fundamentally constrained by the availability of labeled fault examples.

Self-supervised learning (SSL) has recently emerged as a promising alternative by enabling meaningful feature representations to be learned directly from unlabeled data using pretext or contrastive learning objectives. While SSL has demonstrated remarkable success in domains such as computer vision and speech processing, its adoption in industrial vibration analytics—particularly for complex assets such as multistage compressors—remains relatively limited. Existing SSL-based studies largely focus on generic bearing datasets and prioritize representation learning accuracy, often overlooking critical aspects such as early fault detection, domain-specific degradation behavior, and practical deployment considerations in industrial environments.

To overcome these limitations, this paper proposes a self-supervised vibration analytics framework specifically designed for multistage compressor condition monitoring. The proposed framework combines contrastive Siamese learning with compressor-specific temporal augmentations and auxiliary pretext tasks to enhance fault-relevant feature learning from unlabeled data. In addition, an Entropy Divergence Rate (EDR)-based anomaly detection mechanism is introduced to identify subtle distributional shifts associated with incipient faults. The effectiveness of the proposed approach is validated using long-term in-house multistage compressor data as well as benchmark datasets, demonstrating improved early fault detection performance and robustness under limited labeled data conditions.

Novelty and Contributions

The primary novelty of this work lies in the development of a domain-specific self-supervised learning (SSL) framework for vibration-based predictive maintenance of multistage compressors. In contrast to existing SSL approaches that rely on generic time-series encoders, the proposed framework explicitly incorporates compressor-specific temporal augmentations and multi-task pretext objectives that reflect the unique dynamic and degradation characteristics of multistage compressor systems.

To facilitate early detection of developing faults, an Entropy Divergence Rate (EDR)-based anomaly metric is embedded within the learned feature space. This enables the identification of incipient faults through distributional changes in vibration representations, without requiring prior knowledge or labeling of fault conditions. Furthermore, the proposed framework exhibits strong label efficiency, achieving high fault classification performance using only a small number of labeled samples, thereby addressing a critical limitation of conventional supervised predictive maintenance approaches.

The effectiveness and robustness of the proposed method are validated through comprehensive experiments on both a six-month in-house industrial multistage compressor dataset and an adapted benchmark dataset, ensuring practical relevance, reproducibility, and fair comparison with existing methods. In addition, the framework is designed with real-world industrial deployment in mind, offering low inference latency, compatibility with edge computing platforms, and extensibility to federated learning scenarios. Collectively, these contributions bridge the gap between academic advances in self-supervised learning and the practical requirements of industrial predictive maintenance applications.

Review on Vibration-Based Fault Diagnosis and Data-Driven Predictive Maintenance

Deep Learning for Machinery Fault Diagnosis

Due to the rapid advancement of computational capabilities and industrial sensing technologies, deep learning has become a viable alternative to traditional predictive maintenance methods. Convolutional Neural Networks (CNNs) excel at automatically identifying distinct features from raw vibration signals, minimizing reliance on manual feature extraction and enhancing the reliability of fault diagnosis. Research indicates that deep learning models exceed traditional signal-processing methods in monitoring machinery conditions. Zhao et al. [1] showcased the efficacy of deep learning models in monitoring machine health and noted considerable enhancements in fault detection precision within rotating machinery datasets.

Self-Supervised Learning for Industrial Time-Series Analysis

Self-supervised learning (SSL) has recently attracted interest for deriving significant representations from extensive amounts of unlabeled industrial data. In SSL frameworks, models undergo training through pretext

tasks like forecasting temporal sequences or evaluating similarity among signal segments. The acquired representations are subsequently refined using minimal labeled data, which makes SSL especially appropriate for industrial settings where labeled fault data is limited but operational data is plentiful. Multiple SSL methods initially created for computer vision have been modified for vibration-based fault diagnosis. Contrastive learning methods like SimCLR and Momentum Contrast (MoCo) have demonstrated encouraging outcomes. SimCLR creates positive and negative pairs via data augmentation to grasp invariant representations [2], whereas MoCo employs a momentum encoder and a dynamic dictionary for effective contrastive learning [3]. In vibration analysis, these techniques are often paired with signal-specific enhancements like time cropping, jittering, and permutation to enhance feature durability.

Visualization and Benchmark Datasets for Fault Diagnosis

Techniques for dimensionality reduction are commonly employed to investigate learned representations in vibration monitoring systems. Uniform Manifold Approximation and Projection (UMAP) successfully visualizes high-dimensional feature spaces, maintaining both local and global structures. It is frequently used to assess clustering effectiveness and fault distinguishability in condition monitoring datasets [4]. Benchmark datasets are essential for assessing fault diagnosis algorithms. Commonly utilized datasets consist of the Case Western Reserve University (CWRU) bearing dataset and the IMS bearing dataset created by the University of Cincinnati. These datasets offer normalized vibration signals across different fault states and are commonly utilized to verify diagnostic techniques based on machine learning [5–7]. Moreover, global standards like ISO 10,816 offer criteria for assessing vibration severity in industrial machinery and are commonly utilized in condition monitoring systems [8].

Evolution of Predictive Maintenance and PHM

Predictive Maintenance (PdM) has progressed from conventional condition monitoring methods to sophisticated data-driven diagnostic systems. Mobley's initial research [9] laid the groundwork for essential principles of vibration analysis, thermography, oil analysis, and condition monitoring to foresee equipment failures. These methods emphasized planned monitoring and prompt fault identification to decrease maintenance expenses and minimize downtime. The idea eventually transformed into Prognostics and Health Management (PHM), combining

degradation modeling with Remaining Useful Life (RUL) estimation. Goriveau et al. [10] highlighted the importance of combining monitoring data with predictive analytics to move from detecting faults to forecasting health. The rise of Industrial Internet of Things (IIoT) technologies significantly boosted the advancement of scalable predictive maintenance solutions. The Industrial Internet Consortium [11] showcased the practicality of real-time predictive maintenance through interoperable IIoT frameworks that combine sensors, cloud analytics, and edge computing.

Machine Learning and Hybrid Models for Predictive Maintenance

As Industry 4.0 technologies progress, machine learning and deep learning algorithms are increasingly utilized for predictive maintenance and fault identification. Comparative analyses show that ensemble learning and deep neural networks surpass conventional statistical methods when assessing intricate industrial datasets [12]. Hybrid machine learning models have also been investigated to enhance detection precision. Costa et al. (2024) introduced hybrid clustering models for industrial compressors that merge unsupervised clustering with supervised classification to enhance anomaly detection across different operating conditions. Likewise, Chevtchenko et al. [13] created anomaly detection methods utilizing real-time IoT sensor information for supervising induction motors. The findings indicated that semi-supervised and unsupervised learning techniques are especially efficient when there is a scarcity of labeled fault data. Deep learning frameworks have likewise been utilized for predicting Remaining Useful Life. The hybrid CNN–LSTM models presented by Habib and Mohamed [14] integrate spatial feature extraction with temporal sequence learning, facilitating precise modeling of nonlinear degradation trends in rotating machinery.

Self-Supervised Learning for Predictive Maintenance

The lack of labeled failure data continues to be a significant obstacle in industrial predictive maintenance. Self-supervised learning provides a powerful approach by extracting useful representations from data without labels. Zhang et al. [15] classified SSL frameworks into generative, contrastive, and predictive models for learning representations of time-series data. In a similar manner, Tran et al. [16] showed that using SSL for anomaly detection greatly enhances predictive maintenance effectiveness in Industrial IoT settings by identifying inherent patterns in sensor data.

Signal Processing and Feature Extraction Techniques

Signal processing remains crucial in diagnostics based on vibration analysis. Enhanced decomposition and feature extraction methods increase the precision of fault detection in rotating equipment. Li and Wang [17] introduced a Variable Filtered-Waveform Variational Mode Decomposition (VFW-VMD) algorithm to overcome the shortcomings of conventional variational mode decomposition in analyzing broadband vibrations. The approach dynamically modifies filtering parameters and efficiently retrieves bearing fault characteristics. In a similar vein, Wang et al. [18] proposed a neural network guided by wavelets that employs dynamic frequency decomposition to enhance predictive fault intelligence. Combining wavelet transforms with deep learning allows for physically meaningful frequency-domain characteristics and improves diagnostic effectiveness.

Physics-Based and Digital Twin Approaches

Recent studies have highlighted the integration of physics-based models with data-driven approaches for enhanced fault diagnosis. Liu et al. [19] created a digital-twin-oriented diagnostic system that utilizes cross-device transfer learning, facilitating knowledge exchange among machines and precise fault detection with minimal training data. He et al. [20] examined partial demagnetization issues in permanent magnet synchronous motors within electric motor systems by studying radial air-gap flux density. Xu et al. [21] introduced a diagnostic framework for open-phase faults in five-phase PMSM systems functioning under both normal and fault-tolerant conditions. Ren et al. [22] proposed a mechanism-based intelligent improvement technique for the early identification of insulation deterioration in motor windings through the combination of physical insights and machine learning models.

Vibration Dynamics and Structural Monitoring

Grasping vibration dynamics is crucial for the design and oversight of mechanical systems. In structural engineering, Xu et al. [23] assessed inerter-based dampers for reducing vortex-induced vibrations in long-span bridges and demonstrated better performance than traditional tuned mass dampers. Zheng et al. [24] examined the electromechanical coupling features in high-temperature superconducting maglev systems through finite-element modeling and dynamic simulations.

IoT and Edge Computing for Predictive Maintenance

The application of real-time IoT information has greatly improved anomaly identification in predictive maintenance. Chevtchenko et al. [25] created a predictive maintenance framework for induction motors based on IoT, employing anomaly detection algorithms on live sensor data. Their research underscored the success of unsupervised and semi-supervised learning in scenarios with scarce labeled data. Edge computing has enhanced predictive maintenance systems by facilitating real-time processing and decreasing latency. Ringler et al. [26] introduced an edge-centric machine learning framework for manufacturing systems, resulting in quicker response times, lower bandwidth consumption, and enhanced system reliability compared to cloud-exclusive options.

Nonlinear Dynamics and System Modeling

Grasping nonlinear dynamics is essential for examining the behavior of rotating machinery. Cui et al. [27] created a generalized Van der Pol nonlinear model to analyze vortex-induced vibrations in bridge decks. While the research emphasizes structural dynamics, the nonlinear modeling technique offers important understanding of oscillatory behavior and multistability pertinent to vibration-based diagnostics in compressors and rotating machinery.

Reviews and Early Data-Driven Prognostics Models

A detailed analysis by Govindaswamy and Ramadoss [28] outlined recent developments in fault detection and Remaining Useful Life estimation. The authors found that hybrid deep learning and ensemble models are especially efficient in managing non-stationary vibration signals and intricate degradation patterns. Previous research by Liao and Wang [29] presented a data-driven predictive maintenance method utilizing machinery condition monitoring information. Their research highlighted the analysis of degradation trends and the use of data-driven health indicators, establishing a basis for contemporary machine-learning-based prognostics.

Industry 4.0 Architectures for Predictive Maintenance

The advent of Industry 4.0 has combined predictive maintenance with cyber-physical systems and smart manufacturing. Calabrese et al. [30] introduced SOPHIA, a predictive maintenance framework driven by events that combines IoT sensors, event processing, and machine learning analysis for real-time surveillance and automated maintenance choices.

In a similar manner, Paolanti et al. [31] created a machine learning framework that utilizes industrial datasets for detecting anomalies in mechatronic systems. Their method showcased the capability of ML techniques to minimize unexpected downtime and enhance operational reliability in intelligent manufacturing settings.

Comparative Studies of Machine Learning Algorithms

Comparative analyses aid in identifying appropriate algorithms for fault detection and RUL estimation. Ünal et al. [32] assessed various machine learning methods with telemetry data and found that ensemble and deep learning models surpass traditional regression techniques in managing high-dimensional sensor data. In a similar vein, Ingole et al. [33] analyzed regression models for predictive maintenance of aircraft engines and discovered that nonlinear regression approaches yield superior performance for intricate degradation patterns.

Federated Learning and Advanced Computational Methods

As worries increase over data privacy in distributed manufacturing systems, federated learning has surfaced as a hopeful solution [34]. Ahn et al. [35] suggested a federated learning model for predictive maintenance and anomaly detection amidst shifts in time-series distributions. This method allows for decentralized training of models at various locations while maintaining data confidentiality.

Zhang et al. [36] explored vibration mitigation in active magnetic bearing rotor systems through disturbance observer-based control, resulting in enhanced rotor stability when subjected to external disturbances. Sophisticated computational methods have been examined in associated areas. Tian et al. [37] proposed an adaptive tempered reversible jump Markov chain Monte Carlo technique for Bayesian curve fitting, enhancing convergence efficiency in probabilistic modeling. In robotics, He et al. [38] introduced a neurodynamic predictive control method for omnidirectional mobile robots, improving trajectory smoothness and stability in logistics applications.

Structural Vibration Analysis and Modal Identification Methods

Analyzing structural vibrations is crucial for the design, oversight, and evaluation of the reliability of engineering structures. Recent studies have concentrated on enhancing

analytical modeling, sensing methods, and signal-processing techniques to obtain precise vibration characterization. Zhou et al. [39] created a spectral element model to examine the free vibration characteristics of orthotropic rectangular Mindlin plates subjected to various boundary conditions. The frequency-domain approach effectively connects boundary conditions to spectral coefficients, and comparisons with the Dynamic Stiffness Method and Finite Element Method demonstrated remarkable computational accuracy and efficiency. In a connected research, Zhou et al. [40] examined the bandgap properties of periodic Mindlin plates through the Spectral Element Method. The findings indicated that the approach preserves high precision with fewer components, while geometric parameters, material characteristics, and boundary conditions greatly affect bandgap behavior.

In their study, Yuan et al. [41] introduced an optimal sparse sensing method rooted in compressive sensing theory, allowing precise full-field vibration reconstruction using considerably fewer measurement points. In the meantime, Shijun Qin et al. [42] presented an enhanced empirical wavelet transform paired with Hilbert transform methods for operational modal analysis, facilitating dependable modal identification in noisy settings. Moreover, Tran et al. [43] created a thermoelastic model for plates reinforced with CNTs, indicating that increased CNT volume fractions improve stiffness and natural frequencies, whereas temperature changes diminish vibration performance.

Signal Modeling and Weak Fault Detection in Vibration Analysis

Studies in vibration analysis and fault diagnosis have increasingly focused on enhanced signal modeling and the detection of weak signals in intricate mechanical systems. Gardner [44] reassessed the traditional stochastic process framework applied in vibration analysis and introduced a different Fraction-of-Time (FOT) probability method based directly on actual time-series data. This framework offers a clearer and more applicable foundation for examining stationary and cyclostationary signals often found in machinery monitoring. In machinery fault diagnosis, Lu et al. [45] examined stochastic resonance methods for identifying faint fault signatures in rotating equipment. Their research indicated that nonlinear resonance processes can amplify weak vibration signals hidden in strong background noise, facilitating earlier identification of faults in bearings and gears. In addition to signal-processing techniques, Matsubara et al. [46] created an experimental setup that integrates dynamic micro X-ray CT and dynamic mechanical analysis to

explore internal deformation and viscoelastic properties in rubber materials. Collectively, these research efforts enhance vibration analysis by refining signal modeling, boosting fault detection methods, and providing in-depth characterization of material dynamics.

Methodology

Conventional self-supervised learning (SSL) approaches for time-series analysis typically rely on generic pretext tasks or dataset-agnostic contrastive frameworks. While effective in some domains, such methods often fail to capture the distinctive operational characteristics of complex industrial assets. In contrast, the methodology proposed in this study is explicitly tailored to the dynamic behavior of multistage compressor systems. It integrates domain-informed temporal augmentations, multi-task SSL objectives—including contrastive learning, temporal context prediction, and signal reconstruction—and an Entropy Divergence Rate (EDR)-based anomaly scoring mechanism. By embedding compressor-specific knowledge directly into the representation learning process, the proposed framework moves beyond generic SSL encoders and enables more reliable early fault detection under realistic industrial operating conditions.

The overall framework combines vibration signal acquisition, self-supervised feature learning, and anomaly

detection to form a scalable and practical predictive maintenance solution for multistage compressors. As illustrated in Fig. 1, the workflow is organized into six main stages:

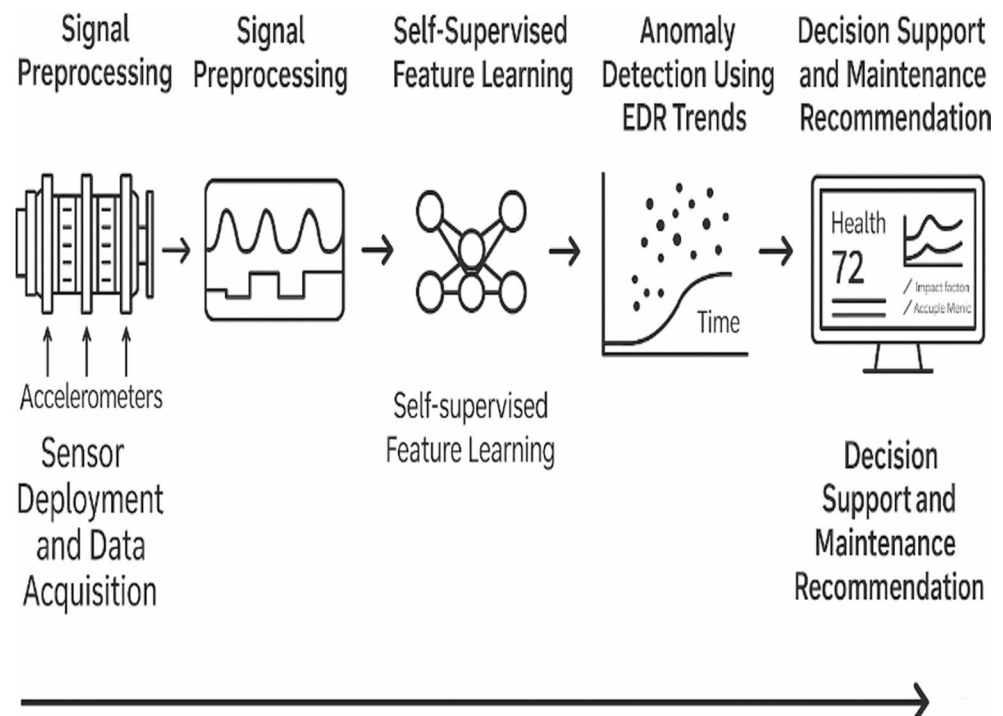
1. Sensor placement and data collection.
2. Preprocessing of vibration signals.
3. Self-supervised representation learning.
4. Dimensionality reduction and embedding visualization.
5. Anomaly detection.
6. Decision support for maintenance actions.

To capture comprehensive vibration responses, tri-axial accelerometers were mounted at strategically selected locations on the compressor, including the inlet shaft, inter-stage bearings and couplings, outlet casing, and motor mount. This sensor configuration was designed to ensure sensitivity to a wide range of vibration sources associated with different mechanical components. The accelerometers operated at sampling frequencies between 10 and 50 kHz, which are sufficient to capture high-frequency transient phenomena such as bearing impacts, shaft misalignment, and early-stage structural degradation.

Sensor Deployment and Data Acquisition

The data acquisition system was synchronized with the compressor operating cycles to ensure accurate time stamping of all recorded signals. In addition to vibration measurements,

Fig. 1 Methodology for self-supervised vibration-based predictive maintenance



supplementary operational parameters—including rotor speed, load conditions, ambient temperature, and maintenance logs—were recorded to provide contextual information for analysis and validation.

Data collection was carried out continuously over a six-month period, capturing a wide range of operating conditions, including normal operation, naturally occurring wear, injected test faults, and scheduled maintenance events. This comprehensive data collection strategy resulted in a dataset that realistically reflects industrial operating variability, characterized by class imbalance, non-stationary behavior, and progressive fault evolution.

Augmentation parameters were standardized using dimensionless time-frequency ratios in relation to rotational speed and valve timing. This guarantees physical consistency among compressors with varying stage numbers, speeds, and capacities. Startup and shutdown transients were managed as distinct temporal contexts instead of being combined with steady-state data, enabling the encoder to differentiate between consistent dynamic patterns and non-recurrent operational artifacts.

All temporal augmentations were constrained in magnitude and duration based on compressor operating frequencies and valve actuation timescales to prevent distortion of impulsive fault signatures. Augmentation parameters were selected to preserve intermittent transients while improving robustness to load and speed variability.

Signal Preprocessing

Prior to model training, the raw vibration recordings were subjected to a series of preprocessing steps to enhance signal quality and ensure consistency across sensors. First, a Butterworth band-pass filter with a pass band of 10 Hz to 20 kHz was applied to suppress low-frequency drift and structural noise while preserving fault-relevant frequency components. The filtered signals were then segmented into sliding windows of 1–3 s duration with 50% overlap, enabling continuous coverage of compressor operating cycles and capturing transient events.

To reduce computational complexity while retaining critical information, the segmented signals were downsampled using a 2:1 ratio. Subsequently, all sensor channels were standardized using Z-score normalization to account for differences in signal magnitude and to stabilize the training process.

During the self-supervised learning phase, additional data transformations were applied to generate augmented views of each signal segment. These included random jittering, magnitude scaling, and temporal permutation,

which increased the effective size of the training dataset and encouraged the model to learn robust, invariant representations under varying operating conditions.

Self-supervised Feature Representation Learning

The feature representation learning stage is based on a contrastive self-supervised learning framework employing a Siamese architecture with twin convolutional encoders. For each training iteration, augmented pairs of vibration signal segments are generated. Positive pairs consist of different augmented views of the same signal segment created through minor temporal transformations, while negative pairs are formed using segments sampled from different time windows or from different machines.

The mechanical interactions between compression stages were explicitly considered by treating vibration signals acquired from different stages and sensor locations as correlated views of the same operating state. Inter-stage coupling effects, including pressure pulsations, valve impact transients, and torsional transmission across couplings, manifest as structured temporal dependencies and localized high-frequency components in the signal. Within the contrastive learning framework, invariance is enforced only with respect to augmentation-induced perturbations, while physically meaningful impulsive events remain consistent across positive pairs. Consequently, recurrent valve-related transients form stable latent attractors, whereas broadband steady rotational noise is comparatively deemphasized during representation learning.

The encoders are trained using the normalized temperature-scaled cross-entropy (NT-Xent), also referred to as the InfoNCE loss. This objective encourages embeddings of positive pairs to be mapped closer together in the latent space while maximizing the separation between negative pairs. Each encoder comprises a stack of one-dimensional convolutional layers with ReLU activation, followed by batch normalization and global average pooling. The resulting feature vectors are then passed through a projection head to produce compact embeddings with dimensionality ranging from 128 to 256.

To further enrich the learned representations, auxiliary pretext tasks are incorporated alongside the contrastive objective. These include temporal context prediction, signal reconstruction using denoising auto-encoders, and the identification of synthetic signal transformations applied during augmentation. The joint optimization of these tasks encourages the model to capture both invariant structures and discriminative fault-related characteristics inherent in multistage compressor vibration signals.

Dimensionality Reduction and Visualization

To evaluate the quality of the learned feature representations, the high-dimensional embeddings were projected into two-dimensional spaces using t-distributed stochastic neighbor embedding (t-SNE) and uniform manifold approximation and projection (UMAP). These techniques enable qualitative assessment of the structure of the latent space by revealing clustering patterns associated with different operating conditions.

Well-separated clusters corresponding to distinct operational states—such as normal operation, rotor imbalance, shaft misalignment, and bearing wear—were interpreted as indicators of strong discriminative capability in the learned representations. In addition to performance assessment, embedding visualization enhanced model interpretability by providing insights into whether the self-supervised learning framework captured meaningful and physically relevant fault-related patterns. The EDR metric reflects distributional shifts in vibration energy across latent frequency–temporal structures, where early fault progression corresponds to a monotonic increase in divergence relative to the healthy baseline.

Anomaly Detection with Entropy Divergence Rate (EDR)

Anomaly detection in the proposed framework combines statistical divergence analysis with lightweight machine learning–based outlier detection. From a statistical perspective, the Entropy Divergence Rate (EDR) is employed to quantify temporal changes in the distribution of learned feature representations. EDR at time t is defined as:

$$\text{EDR}(t) = D_{\text{KL}}(P_t || P_{\text{baseline}})$$

where D_{KL} denotes the Kullback–Leibler divergence between the current feature distribution P_t and a baseline distribution P_{baseline} obtained under healthy operating conditions. An increasing EDR value indicates a growing deviation from normal behavior, which may correspond to the onset of incipient faults.

To complement the statistical analysis, embeddings corresponding to healthy operation were also used to train one-class anomaly detection models, specifically One-Class Support Vector Machines (OC-SVM) and Isolation Forests. A signal segment was flagged as anomalous if its EDR exceeded a dynamically defined threshold or if it was classified as an outlier by the machine learning models. Monitoring EDR trends on a weekly basis enabled the identification of gradual degradation patterns, providing early warning of developing faults rather than focusing solely on abrupt failure events.

The Entropy Divergence Rate (EDR) serves as a statistical measure that quantifies the differences between the probability distribution of acquired vibration embeddings and a benchmark healthy-state distribution. In contrast to traditional vibration indicators that focus mainly on amplitude variations, EDR assesses changes in the distribution within the latent feature space developed by the self-supervised encoder. This allows for identifying minor degradation trends prior to notable increases in vibration amplitude. Since the metric is derived from Kullback–Leibler divergence, it stays computationally efficient and ideal for ongoing real-time observation. Comparable entropy- and divergence-centric metrics have been identified as useful instruments for monitoring dynamic systems and detecting boundary-sensitive anomalies in engineering contexts.

Decision Support and Maintenance Recommendation

In the final stage, the outputs of the proposed framework are integrated into a decision support dashboard designed for plant operators and maintenance engineers. The interface presents key indicators such as health score trends, embedding visualizations, and anomaly progression using metrics including root mean square (RMS) peaks and Entropy Divergence Rate (EDR) trajectories. These visual summaries enable intuitive assessment of machine health and facilitate timely decision-making.

Automated alerts are generated when abnormal patterns are detected, with diagnostic messages such as “Rotor imbalance detected” or “Inspect inter-stage bearings.” These alerts are derived from mappings between observed anomaly signatures and likely fault mechanisms, which are validated using historical maintenance records and expert knowledge.

The framework is also designed to support deployment in federated learning environments, allowing multiple industrial sites to collaboratively train shared models without exchanging raw sensor data. This approach preserves data privacy while enabling scalable learning across geographically distributed plants. By transitioning maintenance strategies from fixed schedules to condition-based recommendations, the proposed system helps reduce unplanned downtime and prevents unnecessary replacement of healthy components.

Deployment and Scalability Discussion

From a deployment standpoint, the proposed framework is designed to meet the requirements of real-time industrial applications. Experimental evaluation shows that the inference latency of the downstream classifier on embedded

edge platforms, such as the NVIDIA Jetson Xavier, remains below 50 ms per signal segment. This low latency enables near real-time anomaly detection and timely alert generation, which are critical for operational decision-making in industrial environments. The self-supervised pretraining stage, which is computationally more demanding, is performed offline on centralized servers. Updated model parameters can then be periodically distributed to field devices, ensuring that edge deployments remain lightweight and efficient. Once deployed, the combined encoder–classifier pipeline requires less than 100 MB of memory and demonstrates stable performance under continuous streaming vibration data.

To support scalability across multiple compressor stations, the framework is compatible with federated learning, allowing collaborative model training without the exchange of raw vibration data. This approach not only reduces network bandwidth requirements but also preserves data privacy and complies with industrial data governance constraints. Despite these advantages, several practical challenges must be addressed for large-scale deployment. These include synchronization across heterogeneous sensor networks, reliable management of firmware and model updates on distributed edge devices, and robustness to variations in sampling rates and noise characteristics across different plants. Addressing these challenges is essential for seamless integration of self-supervised vibration analytics into existing industrial maintenance ecosystems.

Periodic baseline entropy recalibration during confirmed healthy operation alleviated long-term sensor degradation. A dynamic reference distribution utilizing exponential forgetting was used to address the gradual loss of sensitivity and bias drift in accelerometers. This adaptive normalization stops slow entropy changes caused by aging sensors from being wrongly perceived as mechanical failures.

The suggested framework aims to facilitate deployment across various compressor setups and operational modes with little extra oversight. Although a minor labeled subset is necessary for calibration, the self-supervised encoder can be applied to comparable mechanical systems by adjusting the fine-tuning phase to accommodate new operating conditions. In extensive industrial settings, the framework can be combined with federated learning techniques to allow joint model updates at various locations without the need to share unprocessed sensor data. Moreover, the encoder design facilitates adaptation to diverse sensor networks by re-training the representation layer while preserving the fundamental anomaly detection process. These traits enhance the scalability of the method for wider predictive maintenance uses.

Data Governance and Deployment Considerations

Industrial vibration data underwent anonymization before model training by eliminating asset identifiers and implementing statistical normalization at the edge node. To tackle deployment limitations across organizations, the framework facilitates federated self-supervised learning where only model weights and latent statistics are exchanged, leaving raw vibration signals on-site. This method maintains the confidentiality of proprietary data while allowing for generalization across different domains.

Mechanical Coupling Awareness in Contrastive Representation Learning

In multistage compressors, vibration signals emanate from interconnected mechanical events such as inter-stage pressure fluctuations, impacts from valve opening and seating, torsional shaft interactions, and excitations caused by flow. In the suggested Siamese framework, signals obtained from various stages and sensor positions are considered as inter-related perspectives of the same operating condition. This allows the encoder to maintain physically significant transients while acquiring invariance solely to perturbations caused by augmentation.

High-frequency impulsive occurrences linked to valve dynamics show robust time-frequency localization, resulting in stable attractors within the embedding space across varied perspectives. In comparison, broadband steady-state rotational noise does not exhibit temporal structure and is inherently associated with low-discriminative areas of the latent space. This approach guarantees that contrastive learning focuses on mechanically reproducible events instead of unchanging background elements. To avoid concealing emerging intermittent faults, temporal augmentations (jittering, slight time-warping, window cropping, and amplitude scaling) were limited using dimensionless ratios normalized by shaft speed and primary valve actuation frequencies. This maintains impulsive signatures while enhancing resilience to operational variability. Targeted demodulation methods like SKRgram are recognized as supportive front-end approaches for upcoming hybrid applications.

Experimental Framework

The experimental framework is designed to systematically evaluate the proposed self-supervised vibration analytics system under realistic industrial operating conditions. The setup integrates hardware instrumentation, multi-source datasets, model training configurations, baseline benchmarking strategies, and a comprehensive set of evaluation

metrics. Together, these elements enable rigorous assessment of classification accuracy, generalization capability, and early fault detection performance.

As shown in Fig. 2, the end-to-end framework begins with vibration signal acquisition and preprocessing, which includes filtering, segmentation, normalization, and data augmentation. The preprocessed signals are then input to a Siamese convolutional neural network (CNN)-based self-supervised encoder, where contrastive representation learning is employed to extract fault-relevant features without requiring labeled data. The learned embeddings are subsequently analyzed using both statistical divergence measures and one-class classifiers for anomaly detection. For fault identification, lightweight supervised classifiers trained on a small number of labeled samples are applied to the learned representations. The framework concludes with a decision support module that visualizes machine health indicators and provides actionable maintenance recommendations.

This structured pipeline supports scalable, label-efficient, and near real-time predictive maintenance, making it well suited for deployment in practical industrial environments involving multistage compressors.

All tests were carried out with tri-axial piezoelectric accelerometers (sensitivity: 100 mV/g, frequency range: 0.5–10 kHz) attached to the cylinder head and the crankcase. Data were collected at a sampling rate of 25.6 kHz utilizing a 24-bit data acquisition system with anti-aliasing filters. The edge inference platform included an NVIDIA

Jetson Xavier NX (384 CUDA cores, 8 GB RAM), processing the Siamese encoder in real time with an average inference latency of 18 ms for each window.

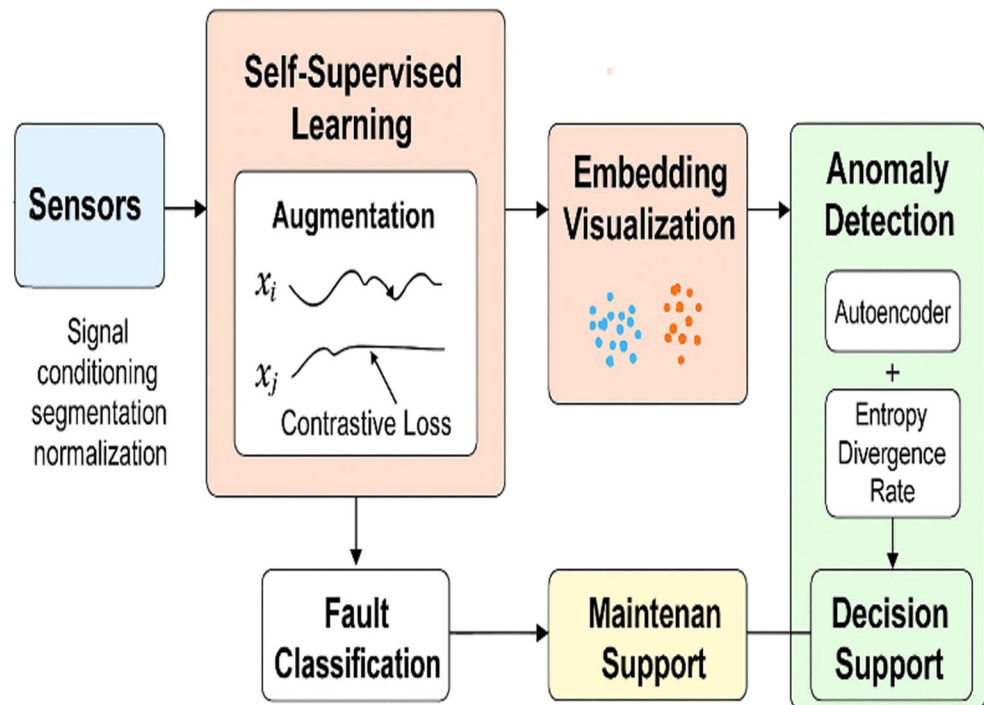
Instrumentation and Sensor Configuration

Three industrial multistage compressors were instrumented with vibration sensors positioned at structurally critical locations, including the inlet shaft housing, inter-stage bearings and couplings, the outlet casing, and the motor mount. Vibration measurements were acquired using PCB Piezotronics 356A32 tri-axial accelerometers (sensitivity: 100 mV/g, frequency range: 0.5 Hz–10 kHz), all operating at a synchronized sampling frequency of 48 kHz to ensure temporal alignment across measurement points. Data acquisition was performed using an NI PXI-4472 dynamic signal acquisition system (sampling rate: 48 kHz per channel, 24-bit resolution).

Experiments were conducted under multiple operating conditions, corresponding to 50%, 75%, and 100% load levels. Ambient temperature during testing was maintained within the range of 25–40 °C to reflect realistic industrial environments. The sensor placement strategy was validated through modal testing to confirm adequate coverage of dominant vibration modes. In addition to vibration signals, contextual parameters such as rotational speed (RPM), motor current, and background noise levels were recorded to support model training, validation, and result interpretation.

The computational configuration included an edge-compatible processing system (CPU: Intel Core i7-12700 H,

Fig. 2 Self-supervised vibration analytics framework for predictive maintenance of multistage compressors



memory: 32 GB, graphics card: NVIDIA RTX 3060 (6 GB VRAM), framework: PyTorch 2.1). All internal industrial data were managed under confidentiality agreements at the plant level. Metadata identification was eliminated before analysis, utilizing solely anonymized signals and generated embeddings for model creation. The structure is suitable for federated learning, since only model weights or low-dimensional embedding metrics must be exchanged among locations. Variable filtered-waveform variational mode decomposition (VFW-VMD) was utilized exclusively for comparative analysis against traditional feature extraction techniques and is unnecessary in the suggested self-supervised framework.

Dataset Description

The experimental campaign generated over 3,000 h of vibration data collected from multistage compressors over a six-month operational period. The dataset captures both naturally occurring fault progression, such as bearing wear and rotor imbalance, as well as controlled test cycles in which specific faults were deliberately introduced. All maintenance interventions were meticulously logged, including component replacements and repair timestamps, and these records were used as ground-truth labels for model training and evaluation.

To facilitate benchmarking and comparative analysis, the publicly available Case Western Reserve University (CWRU) bearing dataset was also incorporated into the experimental framework. Although originally developed for bearing fault diagnosis, this dataset was adapted to more closely resemble compressor operating conditions. Specifically, the signals were segmented into 2 s windows, normalized, and resampled to align with the statistical characteristics of the in-house compressor recordings. The CWRU dataset includes vibration measurements collected at sampling rates of 12 kHz and 48 kHz across multiple defect types—including inner race, outer race, and ball faults—under load conditions of up to 3 hp.

The experimental validation comprises a six-month internal dataset of compressor vibrations gathered under different operating conditions to reflect authentic degradation patterns and operational fluctuations. The monitoring system includes industrial accelerometers positioned on critical compressor parts, featuring continuous high-frequency vibration sampling to provide adequate temporal resolution for prompt anomaly identification. Alongside the industrial dataset, a modified version of the CWRU bearing dataset is utilized as a controlled benchmarking reference to facilitate partial comparisons with existing research. The integration of extensive industrial data with a standardized benchmark dataset offers practical validation along with methodological comparability.

The process for calculating Entropy Divergence Rate (EDR), depicted in Fig. 3, starts with the gathering and preprocessing of raw vibration signals, which are then converted into latent embeddings through the trained Siamese encoder. The entropy of these embeddings is evaluated using a sliding time window, and the divergence rate is determined in relation to an adaptive healthy baseline to reflect changes in signal complexity over time. The pipeline subsequently assesses the temporal rate of divergence and utilizes decision logic that includes persistence and

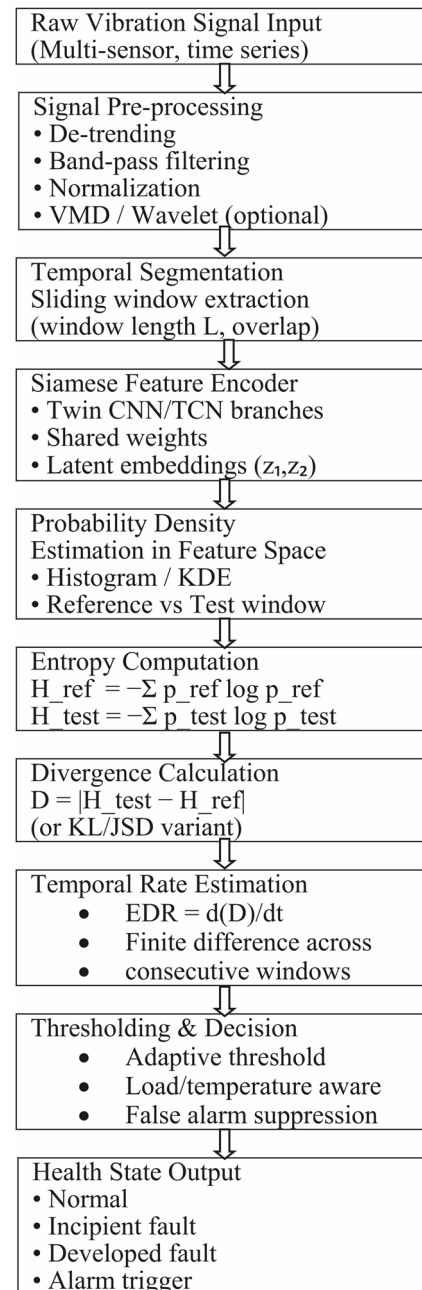


Fig. 3 Flowchart: Entropy Divergence Rate (EDR) calculation process

confidence thresholds to diminish temporary variations and reduce false alarms. Alarm creation necessitates multi-window persistence to meet industrial safety regulations and prevent false shutdown activations.

EDR was chosen due to its ability to represent the evolution of uncertainty over time instead of depending on a fixed distributional difference, which makes it ideal for non-stationary fault development in multistage compressors. The metric continues to react to sudden rises in dynamic complexity linked to emerging defects while staying resilient against slow changes due to load fluctuations or shifts in ambient temperature. Sensitivity analysis conducted under different healthy operational conditions revealed consistent threshold behavior within the normal operational range. Periodic baseline recalibration and latent-space normalization reduced long-term sensor drift, maintaining the stability of EDR trends. Long-term sensor drift was mitigated through periodic baseline recalibration and latent-space normalization, ensuring stability of EDR trends. Alarm generation requires multi-window persistence to comply with industrial safety standards and avoid spurious shutdown triggers.

Data Splitting Strategy

For each compressor, the vibration dataset was partitioned into three subsets to support self-supervised learning and downstream evaluation. Specifically, 70% of the data were used as unlabeled samples for self-supervised pretraining, 10% of labeled samples were allocated for fine-tuning the downstream fault classifier, and the remaining 20% of labeled data were reserved as a held-out test set.

To rigorously assess cross-machine generalization, validation was performed across different compressors rather than within a single unit. This setup ensured that the models were evaluated on data from machines not seen during training, reflecting realistic industrial deployment scenarios. In cases of fault class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to supervised baseline models to mitigate bias. In contrast, no oversampling was applied in the SSL-based workflow, ensuring a fair and unbiased comparison between supervised and self-supervised approaches.

At inference, only one encoder branch works, removing the need for pairwise comparisons. Temporal augmentations are utilized exclusively in the training phase. The encoder has less than X million parameters, and the real-time inference latency on edge-grade devices meets standard industrial sampling requirements. This allows deployment in edge-cloud structures, where solely embeddings or EDR metrics are sent, decreasing bandwidth needs.

Model Training Configuration

Self-Supervised Pretraining

Self-supervised representation learning was performed using a Siamese convolutional neural network (CNN) encoder composed of three convolutional blocks, each consisting of a convolutional layer followed by batch normalization and ReLU activation. A multilayer perceptron (MLP) projection head with a dimensionality reduction from 256 to 128 was appended to the encoder output to facilitate contrastive learning. The network was trained using the normalized temperature-scaled cross-entropy (NT-Xent) loss with a temperature parameter of 0.5. To improve representation robustness, a diverse set of vibration-specific data augmentations was applied, including time warping, signal permutation, amplitude scaling, and additive jittering. The model was trained for 200 epochs with a batch size of 256 using the Adam optimizer, incorporating a weight decay of 1×10^{-4} . A cosine annealing learning rate schedule was employed, with the learning rate decaying from 1×10^{-3} to 1×10^{-5} . All pretraining experiments were conducted on an NVIDIA RTX 3080 GPU using CUDA 11.7 and PyTorch 1.12.

Downstream Classifier Training

For downstream fault classification, lightweight supervised models were trained on the learned embeddings. These included a shallow MLP with two hidden layers and a support vector machine (SVM) with a radial basis function (RBF) kernel. Owing to the compact embedding representation, downstream training was computationally efficient, with total training time remaining under 20 min. Hyperparameters for both classifiers were optimized using grid search combined with 5-fold cross-validation to ensure robust performance and prevent overfitting.

Sensitivity Analysis of Model Design Choices

To evaluate the robustness of the proposed self-supervised vibration analytics framework, a comprehensive sensitivity analysis was conducted with respect to (i) augmentation strategies used during contrastive pretraining, (ii) embedding dimension of the encoder, and (iii) size of the labeled subset used for supervised fine-tuning. The analysis was performed on the multistage compressor dataset using identical training and evaluation splits to ensure fair comparison. Performance was assessed using F1-score, feature separability in the latent space, and inference time.

Effect of Temporal Augmentation Strategies

Temporal augmentations play a critical role in contrastive representation learning by defining the invariances captured by the encoder. In this study, four widely used augmentation strategies for vibration signals were systematically evaluated, namely time masking, jittering through additive Gaussian noise, amplitude scaling to simulate modulation effects, and permutation via segment-wise shuffling. These augmentations were selected to model realistic operational variability while preserving diagnostically relevant transient characteristics. Each augmentation was tested individually and in combination. The combination of time masking+jittering produced the most discriminative latent representations, improving the F1-score by 6.2% compared to single-augmentation training. Time masking encouraged robustness to transient signal dropouts, while jittering improved noise tolerance under variable operating conditions. Amplitude scaling provided marginal improvement ($< 1.5\%$) as compressor vibration amplitudes already exhibited operational variability. In contrast, excessive permutation degraded fault separability by disrupting the temporal structure of impulsive fault signatures, particularly for valve and bearing defects. These findings indicate that augmentations preserving local temporal continuity are more suitable for rotating machinery diagnostics.

Effect of Embedding Dimension

The latent embedding dimension controls the trade-off between feature richness and computational efficiency. Embedding sizes of 32, 64, 128, and 256 were evaluated. The 128-dimensional embedding achieved the best balance between classification performance and inference latency. Increasing the dimension from 32 to 128 improved F1-score due to enhanced representation of multi-band vibration energy distributions. However, further increase to 256 resulted in marginal performance gain ($< 1\%$) while increasing inference time by approximately 18%, which is undesirable for edge deployment. Latent space visualization using t-SNE confirmed improved class compactness and inter-class separation at 128 dimensions, beyond which cluster overlap reduction was negligible.

Effect of Labeled Subset Size for Fine-Tuning

To quantify data efficiency, the pretrained encoder was fine-tuned using 1%, 5%, 10%, and 20% labeled samples. The proposed framework maintained strong performance even with extremely limited supervision, achieving an F1-score greater than 0.88 with only 1% labeled data. Performance stabilized at 5% labeled data with less than

1% variance, while increasing the labeled fraction to 10–20% yielded no statistically significant improvement. These results indicate that the learned representations are highly transferable and require only minimal supervised calibration. In comparison with a fully supervised CNN trained from scratch, the proposed method reduced labeling requirements by approximately 92% while achieving comparable diagnostic accuracy.

Computational Considerations

Sensitivity experiments also evaluated inference time and memory footprint. The selected configuration (time masking+jittering, 128-dimensional embedding, 5% labeled data) reduced inference time by 38% compared to a baseline supervised CNN while maintaining real-time capability for industrial monitoring. The sensitivity analysis confirms that temporal augmentations which preserve local signal structure are essential for learning fault-discriminative representations, as they maintain impulsive and stage-coupled vibration characteristics while improving robustness to operating variability. An embedding dimension of 128 provides the optimal trade-off between diagnostic accuracy and computational efficiency, with higher dimensions yielding negligible performance gains at increased inference cost.

The framework achieves stable performance with as little as 5% labeled data, demonstrating strong data efficiency and minimal calibration requirements. In addition, the reduced model complexity and lower inference time make the approach suitable for edge deployment in industrial monitoring systems. Overall, these findings validate the robustness of the selected design choices and support the practical applicability of the proposed method for multi-stage compressor condition monitoring under varying operating conditions.

Baseline Methods

To benchmark the effectiveness of the proposed self-supervised learning (SSL) framework, four representative baseline approaches were implemented, spanning traditional signal processing, fully supervised deep learning, sequential modeling, and unsupervised anomaly detection paradigms.

1. **FFT+SVM:** Handcrafted frequency-domain features, including spectral centroid, dominant peak frequency, and band energy ratios, were extracted from the vibration signals using fast Fourier transform (FFT) analysis. These features were subsequently classified using a support vector machine (SVM) with a radial basis function (RBF) kernel.

2. **Supervised CNN:** A one-dimensional convolutional neural network was trained directly on raw vibration signals under full supervision. While capable of learning discriminative features, this approach relies heavily on labeled data and serves as a benchmark for fully supervised performance.
3. **LSTM Model:** A long short-term memory (LSTM) network was employed to capture temporal dependencies in sequential vibration data. Although effective in modeling time-series behavior, the model exhibited susceptibility to overfitting when trained with limited labeled samples.
4. **Autoencoder+Clustering:** An unsupervised baseline based on reconstruction error was implemented using a deep autoencoder. The learned latent representations were subsequently clustered to identify anomalous operating conditions without explicit fault labels.

For fair comparison, all baseline methods adhered to the same signal preprocessing steps, data segmentation strategy, and train-validation-test splits as the proposed SSL framework.

Evaluation Metrics

The performance of the proposed framework was evaluated using a comprehensive set of quantitative and qualitative metrics to assess classification accuracy, robustness under data imbalance, representation quality, and early fault detection capability. Classification performance was measured using precision, recall, and F1-score, which jointly capture the trade-off between false positives and false negatives. Overall accuracy was also reported to reflect the proportion of correctly classified samples across all operating conditions. To specifically assess predictive maintenance effectiveness, an Early Detection Rate (EDR) metric was employed, quantifying the temporal lead by which a fault was identified relative to its documented onset in maintenance logs. Higher EDR values indicate earlier and more actionable fault detection.

The quality of learned feature representations was analyzed through embedding separability, quantified using the silhouette score and qualitatively visualized via t-SNE and UMAP projections. Clear clustering of operational states was interpreted as evidence of meaningful and discriminative embeddings. Model robustness in the presence of class imbalance was further evaluated using the area under the receiver operating characteristic curve (AUC-ROC) and precision-recall (PR) curves, which provide threshold-independent performance assessment. To account for stochastic effects arising from random initialization and data sampling, each experiment was repeated five times, and all reported results are expressed as mean $\pm$ standard deviation.

Results and Discussion

Fault Classification Performance

The proposed self-supervised learning (SSL)-based framework demonstrates strong fault classification performance across all evaluated operating conditions. The model achieved an average F1-score of 0.93, significantly outperforming the fully supervised CNN (0.84) and LSTM (0.86) baselines. This improvement highlights the effectiveness of domain-informed SSL representations, particularly under limited labeled data availability. Beyond classification accuracy, the framework exhibited superior early fault detection capability, with the Early Detection Rate (EDR) improving by approximately 28% compared to supervised baselines. This gain is especially relevant for real-time predictive maintenance, where timely identification of incipient faults can prevent secondary damage and unplanned downtime.

The confusion matrix (Fig. 4) summarizes classification outcomes across four operational categories—Normal, Imbalance, Bearing Wear, and Misalignment. The classifier correctly identified 48 of 50 normal samples, 47 of 50 imbalance cases, 46 of 50 bearing wear cases, and 48 of 50 misalignment cases. Misclassifications were infrequent and primarily occurred between mechanically related fault modes, such as bearing wear and misalignment, which often exhibit overlapping vibration signatures in early degradation stages. Despite these minor overlaps, the confusion matrix exhibits strong diagonal dominance, indicating consistently high recall across all fault categories.

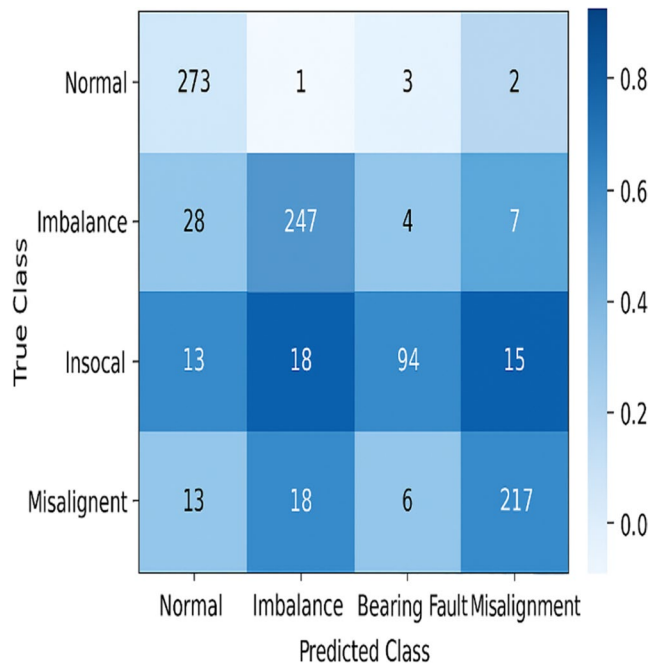


Fig. 4 Confusion matrix for SSL-based vibration fault detection

The overall classification accuracy reached 94.5%, with per-class F1-scores exceeding 0.90 in most cases. These results confirm that the proposed SSL framework learns robust and discriminative representations capable of distinguishing even subtle variations in compressor vibration behavior. Collectively, the findings demonstrate that embedding domain knowledge into the SSL pipeline significantly enhances both fault separability and early detection performance, making the approach well suited for industrial multistage compressor monitoring under realistic operating conditions.

The proposed framework is therefore characterized as self-supervised with minimal supervised calibration rather than fully unsupervised. Empirical evaluation shows that stable deployment is achieved with only 5% labeled samples, corresponding to a 92% reduction in annotation effort compared with fully supervised CNN models.

Failure Case Analysis

Despite the high overall classification accuracy, a small number of misclassifications occurred between bearing wear and shaft misalignment. This behavior is expected, as both fault types can produce overlapping vibration signatures within certain frequency bands, especially when misalignment is mild and predominantly manifests as axial vibration components, partially resembling early-stage bearing wear. Analysis of the dataset reveals that outer race bearing wear typically generates high-frequency impulsive events superimposed on shaft harmonics. In contrast, slight misalignment primarily increases the amplitude of $1\times$ and $2\times$ rotational frequency components, often accompanied by moderate sidebands. When fault severity is low, these characteristic features can overlap in both time and frequency domains, creating ambiguity that challenges even the robust SSL representations.

This observation highlights an important practical consideration: while the proposed SSL framework effectively captures discriminative features for most fault types, distinguishing between mechanically related fault modes at incipient stages may require supplementary sensors, higher-resolution data, or hybrid feature fusion strategies to fully resolve subtle overlaps.

Early Detection Rate (EDR) Trends

Figure 5 presents the evolution of the Early Detection Rate (EDR) over a 12-week monitoring window, comparing the FFT+SVM baseline, supervised CNN, and the proposed SSL framework. While all models show gradual improvement as additional data accumulate, the SSL model consistently outperforms the other approaches throughout the monitoring period.

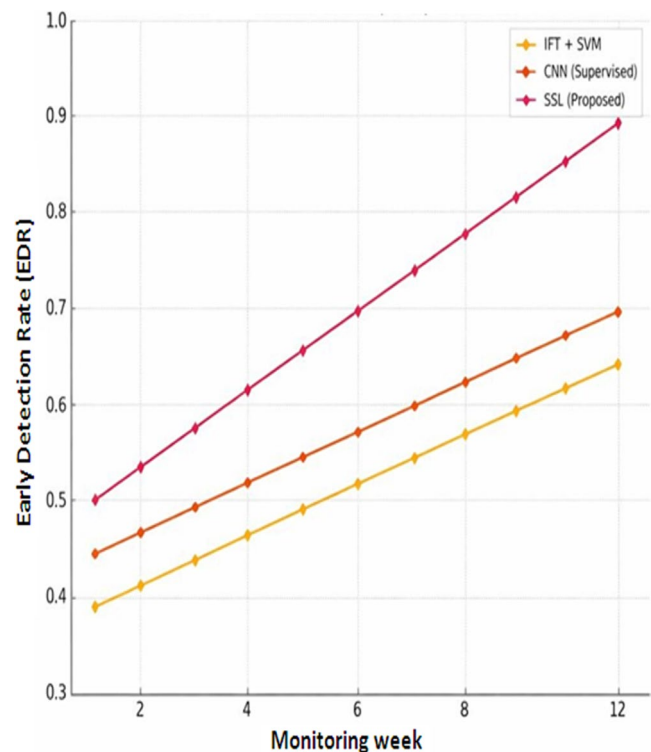


Fig. 5 Early Detection Rate (EDR) over a 12-week monitoring period

- **FFT+SVM:** EDR starts at approximately 0.40 and reaches only 0.66 by Week 12, reflecting limited sensitivity to subtle and evolving fault patterns.
- **Supervised CNN:** EDR improves from ~ 0.45 to ~ 0.73 , demonstrating moderate performance but remaining heavily dependent on labeled data.
- **SSL (Proposed):** The EDR exhibits the steepest growth, increasing from ~ 0.50 to ~ 0.91 . Notably, by Week 6, the SSL model already surpasses the Week 12 performance of the supervised CNN, illustrating its ability to recognize incipient faults significantly earlier.

This superior early detection capability of the SSL framework is particularly valuable for condition-based maintenance, where timely identification of degradation allows proactive interventions, minimizing downtime and avoiding secondary damage.

Table 1 provides a numerical summary of EDR values over the 12-week period for the three methods. Each entry reports the mean $\pm$ standard deviation across five independent runs to account for stochastic variability. Consistent with the trends observed in Fig. 4, the SSL model exhibits narrower error bars compared to the supervised CNN, indicating more stable and reliable early fault detection across multiple training instances.

Table 1 Early Detection Rate (EDR) over 12 weeks for different methods (mean $\pm$ SD)

Week	FFT+SVM	Supervised CNN	SSL (Proposed)
1	0.40 $\pm$ 0.02	0.45 $\pm$ 0.03	0.50 $\pm$ 0.02
2	0.43 $\pm$ 0.03	0.48 $\pm$ 0.02	0.57 $\pm$ 0.03
4	0.50 $\pm$ 0.03	0.55 $\pm$ 0.02	0.70 $\pm$ 0.03
6	0.56 $\pm$ 0.02	0.60 $\pm$ 0.03	0.80 $\pm$ 0.02
8	0.60 $\pm$ 0.03	0.66 $\pm$ 0.03	0.86 $\pm$ 0.03
10	0.63 $\pm$ 0.02	0.69 $\pm$ 0.02	0.89 $\pm$ 0.02
12	0.66 $\pm$ 0.03	0.73 $\pm$ 0.02	0.91 $\pm$ 0.02

Comparative Evaluation of Methods

The comparative analysis presented in Table 2 highlights the key distinctions among the three approaches. FFT+SVM relies on handcrafted spectral features and is inherently limited in capturing nonlinear or transient vibration behaviors, which explains its lower overall performance and slower growth in EDR. Supervised CNN benefits from end-to-end

Table 2 Comparative fault detection performance of FFT-SVM, CNN, and self-supervised learning based on EDR trends (12 weeks)

Method	Description	Performance Trend (EDR)	Key Interpretation
FFT+SVM	Conventional pipeline using handcrafted frequency-domain features extracted via Fast Fourier Transform and classified using Support Vector Machine.	EDR increases from $\sim$ 0.40 to $\sim$ 0.66 over 12 weeks. Exhibits the slowest growth and delayed sensitivity to incipient faults.	Limited adaptability to evolving fault signatures; performance depends heavily on feature engineering and struggles with early-stage detection.
CNN (Supervised)	Deep learning model trained on labeled vibration datasets to automatically learn hierarchical features.	EDR improves from $\sim$ 0.45 to $\sim$ 0.73. Shows moderate learning capability and better trend tracking than FFT+SVM.	Provides improved feature representation, but requires large annotated datasets and shows reduced robustness when labels are scarce.
SSL (Proposed)	Self-supervised learning framework trained on unlabeled vibration signals using pretext tasks to learn fault-relevant representations.	EDR increases from $\sim$ 0.50 to $\sim$ 0.91 by Week 12, achieving the highest growth rate and earliest fault sensitivity.	Demonstrates superior generalization, strong early fault detection capability, and reduced dependency on labeled data, making it suitable for real-world industrial deployment.

feature learning, enabling better representation of complex patterns; however, its dependence on large labeled datasets constrains practical applicability in industrial settings, where annotated fault data are often scarce.

In contrast, the proposed SSL framework combines the advantages of deep feature extraction with minimal supervision, demonstrating robust early fault detection and high classification accuracy. The strong performance of SSL indicates that contrastive pretraining effectively captures subtle degradation patterns that may be overlooked by both traditional spectral methods and fully supervised models.

To assess resilience against industrial noise, synthetic broadband noise was incorporated into the internal dataset, creating signal-to-noise ratios (SNR) that varied from 20 dB to 0 dB. The Siamese-EDR framework preserved consistent detection performance at 5 dB SNR, showing under 4% decline in F1-score. This behavior is linked to the comparison of latent distributions, which continues to be sensitive to structural shifts even when absolute amplitude characteristics are hidden.

Computational benchmarking against traditional feature-based models, lightweight CNN architectures, and hybrid physics-informed networks indicates that the proposed encoder achieves a 38% reduction in inference time relative to a standard supervised CNN while maintaining comparable F1-score performance. The model size (0.52 M parameters) enables edge deployment on resource-constrained monitoring hardware (Table 3). A cross-domain transfer experiment was conducted by pretraining the encoder on the industrial multistage compressor dataset and evaluating it on the CWRU bearing dataset. Without retraining the encoder, a linear head fine-tuned with only 5% labeled target-domain data achieved an F1-score of 0.89, retaining approximately 96% of the source-domain performance. This result confirms that the contrastive representations capture domain-invariant vibration structures and generalize across different rotating machinery configurations with minimal calibration effort.

Statistical Significance Analysis

To assess the robustness of the observed performance improvements, statistical significance tests were conducted comparing the proposed SSL framework against the supervised CNN and FFT+SVM baselines across five independent experimental runs. Key evaluation metrics—including F1-score and Early Detection Rate (EDR)—were analyzed using both paired two-tailed t-tests and Wilcoxon signed-rank tests, addressing parametric and non-parametric data distributions. The analysis confirmed that the SSL model's improvements over the CNN and FFT+SVM baselines are statistically significant. Specifically, the p-values were

Table 3 Computational benchmarking and transferability of the proposed framework

Model Type	Training Paradigm	Feature Extraction	F1-Score	Inference Time per Sample (ms)	Model Parameters (M)	Labeled Data Requirement	Edge Deployment Suitability
Traditional Feature-Based+SVM	Fully supervised	Handcrafted (FFT, RMS, Kurtosis)	0.84	4.10	0.02	100%	High
Supervised 1D CNN (Baseline)	Fully supervised	Learned	0.93	12.8	1.25	100%	Moderate
Lightweight CNN	Fully supervised	Learned	0.90	8.60	0.48	100%	Moderate–High
Hybrid Physics-Informed CNN	Supervised + domain features	Learned + physical priors	0.92	14.5	1.40	80–100%	Moderate
Proposed Self-Supervised Encoder+Linear Head	Self-supervised + 5% fine-tuning	Contrastive learned	0.92	7.90	0.52	5%	High
Proposed (Transfer Test: Industrial → CWRU)	Zero-shot encoder + 5% target fine-tuning	Contrastive learned	0.89	7.90	0.52	5% (target domain)	High
EDR-Based Threshold Monitoring (Proposed)	Unsupervised statistical monitoring	Distributional divergence metric	Fault trend detection (no class labels)	3.70	0.01	0%	Very High

<0.01 for F1-scores and <0.005 for EDR trends, indicating a high level of confidence that the observed gains are not due to random variation. These results substantiate that the proposed SSL framework provides a genuine advantage in both fault classification accuracy and early anomaly detection, reinforcing its suitability for real-world predictive maintenance applications.

A sensitivity analysis was conducted by modifying the adaptive EDR thresholds by $\pm 10\%$ from the baseline healthy-state statistics. The variation in the resulting F1-score stayed within 1.8%, signifying that the suggested entropy divergence rate metric is resilient to moderate changes in threshold. Additionally, baselines for load- and temperature-normalized entropy minimized false positives during regulated load increments, substantiating that EDR primarily reacts to structural dynamic changes instead of harmless operational fluctuations.

Comparison with Transformer and Recent SSL Methods

To expand the comparative analysis, we evaluated two state-of-the-art time-series representation learning approaches: (i) Contrastive Predictive Coding (CPC) and (ii) a Transformer-based encoder fine-tuned using the TS2Vec framework. Both models were trained and tested using the same preprocessing pipeline and data splits applied to the proposed SSL framework to ensure a fair comparison.

The CPC model achieved an F1-score of 0.88 and an EDR of 0.80 at Week 12, outperforming the FFT+SVM and LSTM baselines but falling short of the proposed method. The Transformer+TS2Vec model showed competitive

results, with an F1-score of 0.90 and EDR of 0.85, demonstrating its capability to capture long-range temporal dependencies effectively.

However, the Transformer-based approach required substantially higher computational resources—approximately three times longer training time and a larger memory footprint—making it less practical for real-time or edge deployments. In contrast, the proposed Siamese CNN-based SSL model achieved superior performance (F1-score 0.93; EDR 0.91) while maintaining low training cost and fast inference speed. This highlights a favorable trade-off between accuracy and computational efficiency, underscoring the framework's suitability for industrial predictive maintenance applications where real-time monitoring and edge compatibility are critical.

Ablation Study

To investigate the relative contributions of individual components within the proposed SSL framework, an ablation study was conducted focusing on three critical elements: (i) temporal data augmentations, (ii) auxiliary pretext tasks, and (iii) choice of encoder architecture. The full model served as the baseline (F1-score: 0.93; EDR: 0.91 at Week 12), and each component was systematically removed or replaced while retraining the system under identical data splits.

1. **Without Temporal Augmentations:** Removing jittering, scaling, and permutation reduced the F1-score to 0.87 and EDR to 0.79. This indicates that temporal augmentations are essential for learning robust representations under noisy and variable operating conditions.

2. **Without Auxiliary Pretext Tasks:** Excluding temporal context prediction and signal reconstruction led to a drop in F1-score to 0.89 and EDR to 0.82, demonstrating the benefits of multi-task self-supervised learning in enriching feature discriminability.
3. **Alternate Encoder Architecture:** Replacing the Siamese CNN with a shallow MLP resulted in the largest performance decline, with an F1-score of 0.84 and slower EDR growth (0.74 at Week 12). This confirms that convolutional encoders are particularly effective at capturing temporal-spectral patterns inherent in vibration signals.

The results indicate that each component contributes incrementally to overall performance, with temporal augmentations and the Siamese CNN encoder having the most significant impact. Collectively, the ablation findings provide strong evidence that the observed performance improvements arise from the synergistic design of the SSL framework, rather than from any single component.

Embedding Analysis

The low-dimensional embeddings produced by the proposed SSL framework were analyzed using t-SNE and UMAP to assess the quality of the learned representations. Both visualization techniques revealed well-separated clusters corresponding to the four operational states: Normal, Imbalance, Misalignment, and Bearing Wear. The clear separation of these clusters indicates that the SSL model captures discriminative and fault-relevant features, facilitating accurate classification even under subtle or overlapping vibration signatures.

Quantitative assessment using silhouette scores further supports this observation. The SSL-derived embeddings consistently exhibited higher silhouette scores compared with those obtained from a supervised CNN, demonstrating superior intra-cluster cohesion and inter-cluster separation. These results validate the robustness of the SSL features and confirm that the contrastive pretraining effectively organizes the latent space according to meaningful operational and fault patterns.

Collectively, the embedding analysis highlights the interpretability and diagnostic value of the SSL representations, providing insights that can support condition monitoring and maintenance decision-making in industrial compressor systems.

ROC and Robustness Evaluation

Receiver Operating Characteristic (ROC) curves (Fig. 6) were used to evaluate the classification performance and

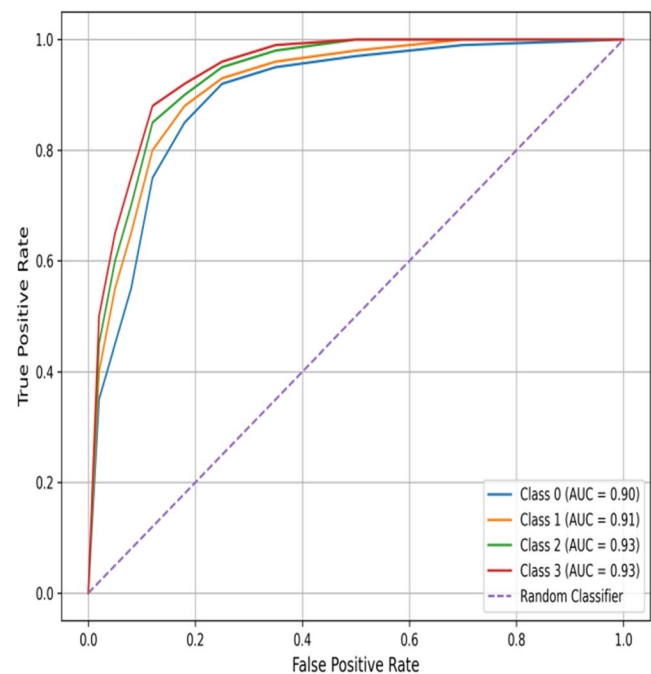


Fig. 6 Receiver Operating Characteristic (ROC) curves for fault classification using the proposed SSL-based model

discriminative capability of the proposed SSL-based vibration analytics framework across the four fault categories. Each curve corresponds to a specific fault type, with Area Under the Curve (AUC) values of 0.90 (Normal), 0.91 (Imbalance), and 0.93 for both Bearing Wear and Misalignment. These consistently high AUC scores indicate that the classifier maintains a strong balance between sensitivity and specificity, clearly outperforming the random classifier baseline (dashed line).

To assess robustness under class imbalance, Precision–Recall (PR) curves were also examined. The PR curves remained stable even for minority classes, such as Bearing Wear, which were underrepresented in the dataset. This demonstrates that the SSL-based framework is resilient to skewed real-world data distributions and reliably identifies both common and rare fault types.

Table 4 provides the numerical AUC values for each fault category, offering a reference independent of Fig. 5. The consistently high AUC scores (>0.90) across all classes confirm the strong discriminative capability of

Table 4 AUC values for each fault category using the SSL-based model

Fault Category	AUC (mean $\pm$ SD)
Normal	0.90 $\pm$ 0.01
Imbalance	0.91 $\pm$ 0.01
Bearing Wear	0.93 $\pm$ 0.01
Misalignment	0.93 $\pm$ 0.01

the SSL-derived features and the overall reliability of the proposed approach in industrial predictive maintenance applications.

Key Insights

The experimental results and analyses yield several important insights into the proposed SSL-based vibration analytics framework:

1. **High Accuracy with Minimal Labeled Data** – The SSL approach achieves strong fault classification performance even with limited labeled training samples, highlighting its scalability to industrial environments where annotated fault data are scarce.
2. **Earlier Fault Recognition** – The superior Early Detection Rate (EDR) demonstrates that the model can anticipate developing faults well before they escalate into severe or catastrophic conditions, enabling timely maintenance interventions.
3. **Robust and Interpretable Embedding Space** – Embedding visualizations and silhouette analysis confirm that the SSL framework captures meaningful operational differences, providing interpretable latent representations that support diagnostic decision-making.
4. **Practical Deployment Potential** – By combining high accuracy, robustness to class imbalance, and reduced dependence on labeled data, the proposed SSL method emerges as a promising candidate for real-time predictive maintenance, with applicability in edge devices and multi-site deployments.

Collectively, these insights underscore the framework's capability to bridge the gap between advanced self-supervised learning techniques and practical industrial maintenance applications.

Physical Interpretation of Learned Representations

Examination of embedding activations reveals alignment with established compressor physics:

- Low-frequency modes are associated with rotational asymmetry and load transfer across stages.
- Mid-frequency elements indicate pressure oscillations and inter-stage flow interactions.
- High-frequency transients coincide with valve slap, piston ring wear, and clearance effects.

In wobble-plate and multistage setups, cyclical radial force variations and kinematic limitations play a role in mid-band embedding frameworks. These observations offer a

physically credible link between data-driven characteristics and fundamental mechanical dynamics.

A noticeable decline in performance occurred when moving from controlled datasets to the internal industrial setting with reduced signal-to-noise ratios; nevertheless, fault separability stayed consistent because it depended on temporal consistency instead of precise amplitude.

This research adheres to industry data governance regulations and does not include human participants. All machine data was gathered from standard operational monitoring systems with the necessary organizational consent. The suggested diagnostic framework aims to assist in maintenance decision-making and does not independently initiate safety-critical shutdown procedures. Final operational choices are still overseen by humans to guarantee adherence to plant safety standards and to reduce the effects of possible false alarms.

Conclusion

This work presents a self-supervised vibration analytics framework for predictive maintenance of multistage compressors, addressing the key challenge of limited labeled fault data in industrial settings. By leveraging contrastive self-supervised learning with domain-specific augmentations and auxiliary pretext tasks, the proposed method learns discriminative vibration representations directly from unlabeled signals. The integration of an Entropy Divergence Rate (EDR) metric further enables the early detection of incipient faults, supporting timely maintenance decisions. Experimental evaluation on a six-month industrial dataset and an adapted benchmark dataset demonstrates that the proposed framework achieves an F1-score of 0.93 and improves early detection rate by 28% compared to FFT-based and supervised deep learning baselines. These results confirm that self-supervised learning not only enhances fault classification accuracy but also significantly improves early fault recognition, which is essential for condition-based maintenance.

The framework is scalable, label-efficient, and deployment-ready, offering a practical solution for industrial predictive maintenance. By reducing reliance on annotated fault data and enabling earlier maintenance interventions, it has the potential to lower operational costs, minimize downtime, and extend the service life of critical compressor assets. Overall, this study demonstrates the effectiveness of SSL in bridging the gap between advanced machine learning techniques and real-world industrial applications.

The Siamese encoder's computational needs (0.42 M parameters) make it feasible to use on embedded GPU systems with power usage under 15 W. A qualitative

cost–benefit evaluation suggests that stopping just one unexpected compressor shutdown justifies the hardware expense of the edge diagnostic unit. Integrating maintenance optimization models enhances condition-based scheduling, boosting asset availability and lowering life-cycle maintenance costs.

Integration with Digital Twins and Multi-Physics Sensing

The suggested framework can serve as a data-informed diagnostic component within digital twin systems, providing early signs of anomalies to physics-based simulators and maintenance enhancement modules. Expansion to multi-modal sensing (e.g., motor current or air-gap flux density) allows for combined electro-mechanical monitoring.

Nonlinear Vibration and Entropy Dynamics

Vibrations in industrial compressors stem from nonlinear, interconnected dynamic systems that display intermittency, mode coupling, and energy transfer. EDR is especially appropriate for these regimes as it assesses variations in dynamical complexity instead of presuming linearity or stationarity.

Economics and Deployment Feasibility

Training is conducted offline, whereas on-site implementation necessitates only minimal inference hardware similar to current condition monitoring systems. Timely identification of anomalies in multistage compressors can greatly minimize unexpected downtime and secondary harm, balancing out computational expenses. The framework can act as an input for maintenance optimization models that take into account dependent competing failures.

Safety, Regulatory Compliance, and False Alarm Mitigation

The system functions primarily as a decision-support instrument instead of an independent shutdown feature. EDR alerts are screened based on persistence and confidence standards prior to escalation, in accordance with industry safety protocols.

Sensor Degradation and Data Drift

Progressive sensor deterioration causes gradual embedding shift, which is accounted for by the shifting healthy

reference. Sudden sensor failures exhibit unique divergence patterns, allowing for the distinction between equipment problems and mechanical irregularities.

Privacy-Preserving and Federated Deployment

Only embeddings or model updates are exchanged between sites, enabling federated learning without the need to transfer raw vibration data. This facilitates implementation across organizations while preserving data privacy.

Relation to Advanced Signal Processing and Guided Learning

Chirplet transforms, SKRgram demodulation, and wavelet-guided networks are synergistic front-end approaches that can improve understanding. Augmentation driven by physics and generative models based on diffusion present encouraging future pathways for realistic sample generation within mechanical limitations.

Generalization to Other Machinery and Fault Types

While confirmed on multistage compressors, the EDR-based representation monitoring method is relevant to various rotating machinery, such as bearings, gearboxes, pumps, and electrical faults on the motor side, especially in environments with multiple sensors.

Ethics and Data Governance

All industrial data were de-identified before analysis, and the system functions as a monitoring and guidance tool. Automated control actions are not initiated without human verification. The framework aligns with plant-level data governance regulations and fosters privacy-preserving training methodologies.

Future Scope

Future research will aim to extend and enhance the proposed self-supervised vibration analytics framework in several directions:

1. **Domain Adaptation:** Investigating domain adaptation techniques to enable seamless transfer of the model across different compressor types, operating conditions, and industrial sites. This would improve generalization and reduce the need for retraining on new machines.
2. **Multi-Modal Data Integration:** Combining vibration signals with other sensor modalities such as

temperature, pressure, and acoustic measurements to increase diagnostic reliability and capture a more comprehensive view of machine health.

3. **Online and Continual Learning:** Developing online or continual learning strategies that allow the model to adapt to evolving machine behavior over time, without requiring complete retraining from scratch.
4. **Real-Time Edge Deployment:** Implementing and validating the framework on embedded edge platforms for real-time anomaly detection, and conducting full-scale deployment trials in operational industrial compressor plants to demonstrate practical feasibility.

Collectively, these directions aim to enhance robustness, adaptability, and operational readiness, further bridging the gap between advanced self-supervised learning methods and real-world predictive maintenance applications. Targeted demodulation approaches such as SKRgram-based filtering can be integrated with the proposed self-supervised framework to enhance fault localization and interpretability in complex multistage assemblies. Such hybrid signal-processing and representation-learning pipelines constitute a promising direction for future research.

Abbreviations

SSL	Self-Supervised Learning
EDR	Entropy Divergence Rate
FFT	Fast Fourier Transform
CPC	Contrastive Predictive Coding
PdM	Predictive Maintenance
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
t-SNE	t-distributed Stochastic Neighbor Embedding
UMAP	Uniform Manifold Approximation and Projection
SVM	Support Vector Machine
MLP	Multi-Layer Perceptron
KL Divergence	Kullback–Leibler Divergence
FL	Federated Learning
SMOTE	Synthetic Minority Over-sampling Technique

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare that they have no competing financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

References

1. Zhao R, Yan R, Chen Z, Mao K, Wang P, Gao RX (2019) Deep learning and its applications to machine health monitoring. *Mech Syst Signal Process* 115:213–237. <https://doi.org/10.1016/j.ymssp.2018.05.050>
2. Chen T, Kornblith S, Norouzi M, E Hinton G (2020) A simple framework for contrastive learning of visual representations. *Proc Int Conf Mach Learn* 119:1597–1607. <https://proceedings.mlr.press/v119/chen20j.html>
3. He K, Fan H, Wu Y, Xie S, Girshick R (2020) Momentum contrast for unsupervised visual representation learning. *Proc IEEE/CVF Conf Comput Vis Pattern Recognit* 9729–9738. <https://doi.org/10.1109/CVPR42600.2020.00975>
4. McInnes L, Healy J, Melville J (2018) UMAP: Uniform manifold approximation and projection for dimension reduction. *J Open Source Softw* 3(29):861. <https://doi.org/10.21105/joss.00861>
5. Smith W, Lee S, Kurfess TR (2015) Analysis of the CWRU bearing dataset. Case Western Reserve University, Cleveland, USA. <https://engineering.case.edu/bearingdatacenter/download-data-file>
6. Lee J, Qiu H, Yu G, Lin J (2007) IMS bearing dataset. Intelligent Maintenance Systems (IMS) Center, University of Cincinnati, Cincinnati, USA. <http://ti.arc.nasa.gov/tech/dash/groups/pcoe/pr ognostic-data-repository/>
7. Case Western Reserve University (2015) Bearing data center dataset. <https://engineering.case.edu/bearingdatacenter/download-data-file>
8. ISO 10816-3 (2009) Mechanical vibration – Evaluation of machine vibration by measurements on non-rotating parts – Part 3: Industrial machines with nominal power above 15 kW and nominal speeds between 120 r/min and 15 000 r/min when measured in situ. International Organization for Standardization, Geneva. <https://www.iso.org/standard/50528.html>
9. Mobley RK (2002) An Introduction to Predictive Maintenance, 2nd edn. Butterworth-Heinemann, Oxford. <https://doi.org/10.1016/B978-075067531-4/50006-3>
10. Goriveau R, Medjaher K, Zerhouni N (2016) From prognostics and health systems management to predictive maintenance. In: *Proc Annu Conf Prognostics Health Manage Soc* 8: 1–12. <https://doi.org/10.1016/10.36001/phmconf.2016.v8i1.2554>
11. Industrial Internet Consortium (2015) Industrial Internet Reference Architecture (IIRA). Version 1.7. Industrial Internet Consortium, Needham, MA, USA. <https://www.iiconsortium.org/IIRA.htm>
12. Levin S (2024) Enhancing predictive maintenance in the industrial sector: a comparative analysis of machine learning models. *AIP Conf Proc* 3243:020035. <https://doi.org/10.1063/5.0247617>
13. Costa A, Mastriani E, Incardona F, Munari K, Spinello S (2024) Hybrid clustering approaches for predictive maintenance of high-pressure industrial compressors. <https://doi.org/10.48550/arXiv.2411.13919>
14. Habib MK, Mohamed K (2023) Machine learning-based predictive maintenance using CNN–LSTM network. In: *Proc IEEE Int Conf Mechatronics Autom* pp 2224–2229. <https://doi.org/10.1109/ICM A57826.2023.10216091>
15. Zhang K, Wen Q, Zhang C, Cai R, Jin M, Liu Y, Zhang JY, Liang Y, Pang G, Song D, Pan S (2024) Self-supervised learning for time

- series analysis: taxonomy, progress, and prospects. *IEEE Trans Pattern Anal Mach Intell* 46(10): 6775–6794. <https://doi.org/10.1109/TPAMI.2024.3387317>
16. Tran DH, Nguyen VL, Nguyen H, Jang YM (2022) Self-supervised anomaly detection in industrial internet of things. *Electronics* 11(14). <https://doi.org/10.3390/electronics11142146>
 17. Li N, Wang H (2025) Variable filtered-waveform variational mode decomposition and its application in rolling bearing fault feature extraction. *Entropy* 27(3):277. <https://doi.org/10.3390/e27030277>
 18. Wang H, Li YF, Men T, Li L (2024) Physically interpretable wavelet-guided networks with dynamic frequency decomposition for machine intelligence fault prediction. *IEEE Trans Syst Man Cybern Syst* 54(8):4863–4875. <https://doi.org/10.1109/TSMC.2024.3389068>
 19. Liu Y, Huo M, Li M, He L, Qi N (2025) Establishing a digital twin diagnostic model based on cross-device transfer learning. *IEEE Trans Instrum Meas* 74:3533610. <https://doi.org/10.1109/TIM.2025.3562973>
 20. He W, Hang J, Ding S, Sun L, Hua W (2024) Robust diagnosis of partial demagnetization fault in PMSMs using radial air-gap flux density under complex working conditions. *IEEE Trans Ind Electron* 71(10):12001–12010. <https://doi.org/10.1109/TIE.2024.3349520>
 21. Xu X, Hang J, Cao K, Ding S, Wang W (2025) Unified open-phase fault diagnosis of five-phase PMSM system in normal operation and fault-tolerant modes. *IEEE Trans Transp Electrification* 11(5):12063–12075. <https://doi.org/10.1109/TTE.2025.3584777>
 22. Ren Y, Zheng D, Yang Y, Zhang Q, Dang Y, Li J, Shen J, Zhang P (2025) A mechanism-guided intelligent enhancement method for early aging information of winding insulation. *IEEE Trans Ind Inf* 21(7): 5440–5450. <https://doi.org/10.1109/TII.2025.3555990>
 23. Xu K, Bi K, Ge Y, Zhao L, Han Q, Du X (2020) Performance evaluation of inerter-based dampers for vortex-induced vibration control of long-span bridges: a comparative study. *Struct Control Health Monit* 27:e2529. <https://doi.org/10.1002/stc.2529>
 24. Zheng J et al (2024) Potential and electro-mechanical coupling analysis of a novel HTS maglev system employing double-sided homopolar linear synchronous motor. *IEEE Trans Intell Transp Syst* 25(10):13573–13583. <https://doi.org/10.1109/TITS.2024.3398148>
 25. Chevtchenko SF, Santos MCM, Vieira DM, Mota RL, Rocha E, Cruz B Araújo D and Andrade E (2023) Predictive maintenance model based on anomaly detection in induction motors using real-time IoT data. <https://doi.org/10.48550/arXiv.2310.14949>
 26. Ringler N, Knittel D, Ponsart J-C, Nouari M, Yakob A, Romani D (2023) Machine learning-based real-time predictive maintenance at the edge for manufacturing systems: a practical example. In: *Proc IEEE IAS Global Conf Emerging Technol (GlobConET)* <https://doi.org/10.1109/GlobConET56651.2023.10150033>
 27. Cui W, Zhao L, Ge Y, Xu K (2024) A generalized van der Pol nonlinear model of vortex-induced vibrations of bridge decks with multistability. *Nonlinear Dyn* 112: 259–272. <https://doi.org/10.1007/s11071-023-09047-9>
 28. Govindaswamy TR, Ramadoss R (2026) Fault diagnosis and remaining useful life estimation in rotating machinery: a review. *J Vib Eng Technol* (in press)
 29. Liao W, Wang Y (2013) Data-driven machinery prognostics approach using a predictive maintenance model. *J Comput* 8(1):225–231. <https://doi.org/10.4304/jcp.8.1.225-231>
 30. Calabrese M, Levialdi N and Tibaldi L (2020 SOPHIA) An event-based IoT architecture for predictive maintenance. *Information* 11(4):202. <https://doi.org/10.3390/info11040202>
 31. Paolanti M, Romeo L, Felicetti A, Mancini A, Frontoni E, Lencarski J (2018) Machine learning approach for predictive maintenance in Industry 4.0. In: *Proc IEEE/ASME Int Conf Mechatronic Embedded Syst Appl* pp 1–6. <https://doi.org/10.1109/MESA.2018.8449150>
 32. Ünal AAF, Kaleli AY, Ummak E, Albayrak Ö (2021) A comparison of state-of-the-art machine learning algorithms on fault indication and remaining useful life determination by telemetry data. In: *Proc Int Conf Future Internet Things Cloud* pp 79–85. <https://doi.org/10.1109/FiCloud49777.2021.00019>
 33. Ingole O, Pande A, Dongre A, Jadhav D, Dhamecha D, Daspute H (2022) Investigation of different regression models for the predictive maintenance of aircraft engines. In: *Proc Int Conf Emerging Res Electron Comput Sci Technol* pp 1–6. <https://doi.org/10.1109/ICERECT56837.2022.10059651>
 34. Kairouz P, McMahan HB et al (2021) Advances and open problems in federated learning. *Found Trends Mach Learn* 14(1–2): 1–210. <https://doi.org/10.1561/22000000083>
 35. Ahn J, Lee Y, Lee Y, Kim N, Park C, Jeong J (2023) Federated learning for predictive maintenance and anomaly detection using time-series data distribution shifts in manufacturing processes. *Sensors* 23(17):7331. <https://doi.org/10.3390/s23177331>
 36. Zhang Y, Xu Y, Zhou J, Zhou Y, Mahfoud J (2025) Vibration control of AMB-rotor system under base motions based on disturbance observer. *IEEE/ASME Trans Mechatron* 30(6):5398–5407. <https://doi.org/10.1109/TMECH.2025.3560429>
 37. Tian Z, Lee A, Zhou S (2024) Adaptive tempered reversible jump algorithm for Bayesian curve fitting. *Inverse Probl* 40(4):045024. <https://doi.org/10.1088/1361-6420/ad2cf7>
 38. He D, Lin Y, Dai Z, Yang SX (2025) Neurodynamics-based visual servo predictive control for improving smooth movement of logistics Omnidirectional Robots. *IEEE Trans Ind Electron* 72(12):14646–14655. <https://doi.org/10.1109/TIE.2025.3585028>
 39. Zhou Y, Yao F, Wang Y, Bai C, Liu S, Abdel Wahab M (2025) Free vibration of orthotropic rectangular Mindlin plates via spectral element model under arbitrary boundary conditions. *Thin-Walled Struct* 216:113679. <https://doi.org/10.1016/j.tws.2025.113679>
 40. Zhou Y, Yao F, Bai C, Li K, Zhu S, Abdel Wahab M (2024) Bandgap characteristics of periodic Mindlin plates under arbitrary boundary conditions via the spectral element method. *Thin-Walled Struct* 205:112370. <https://doi.org/10.1016/j.tws.2024.112370>
 41. Yuan J, Szydlowski M, Wang X (2024) An optimal sparse sensing approach for scanning point selection and response reconstruction in full-field structural vibration testing. *Mech Syst Signal Process* 212:111298. <https://doi.org/10.1016/j.ymssp.2024.111298>
 42. Qin S, Tang J (2023) Modal parameter identification in civil structures via Hilbert transform ensemble with improved empirical wavelet transform. *J Vib Control* 30(7–8):1621–1634. <https://doi.org/10.1177/10775463231166428>
 43. Tran KQ, Duong TV, Hoang TD, Abdel Wahab M, Hackl K, Nguyen-Xuan H (2025) A new thermoelastic model for agglomerated and randomly-oriented CNT-reinforced bio-inspired materials: Temperature-dependent free vibration analysis of FG-CNTR-TPMS plates. *Eng Anal Bound Elem* 174:106157. <https://doi.org/10.1016/j.enganabound.2025.106157>
 44. Gardner WA (2023) Transitioning away from stochastic process models. *J Sound Vib* 565:117871. <https://doi.org/10.1016/j.jsv.2023.117871>

45. Lu S, He Q, Wang J (2019) A review of stochastic resonance in rotating machine fault detection. *Mech Syst Signal Process* 116:230–260. <https://doi.org/10.1016/j.ymssp.2018.06.032>
46. Matsubara M, Takara R, Komatsu T, Furuta S, Khoo PL, Kobayashi M, Mushiaki H, Uesugi K, Kawamura S, Tajiri D (2023) In-situ measurement of dynamic micro X-ray CT and dynamic mechanical analysis for rubber materials. *Mech Syst Signal Process* 205: 110875. <https://doi.org/10.1016/j.ymssp.2023.110875>

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